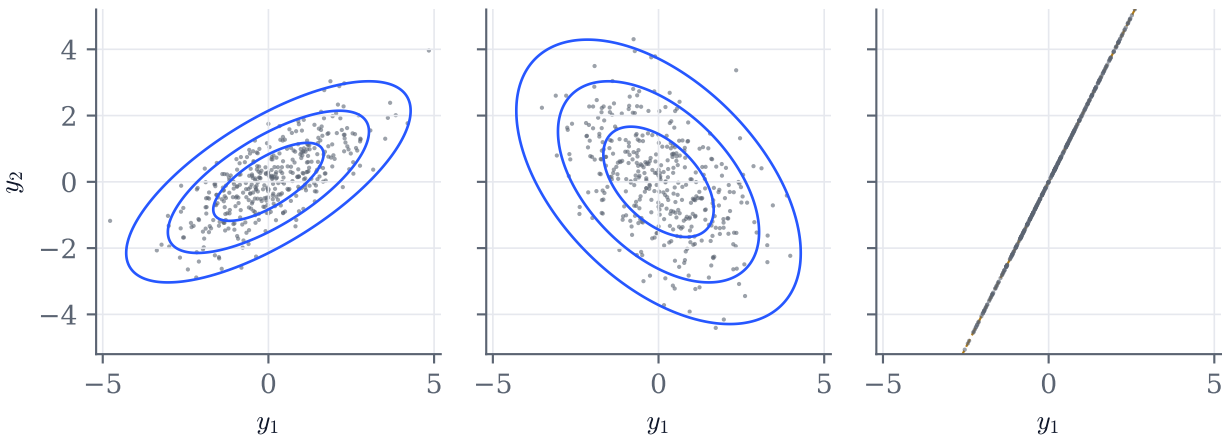

invertible $C$, $\det C = 1$
invertible $C$, $\det C = 3$
singular $C$: degenerate
$y_2$
$y_1$
$y_1$
$y_1$
4
2
0
−2
−4
−5
0
5
sample of 400 points
ellipses holding 50%, 90%, 99% of the law
the line $y_2 = 2y_1$